ECON 770 Recitation 6

Sept 18, 2020

University of North Carolina at Chapel Hill

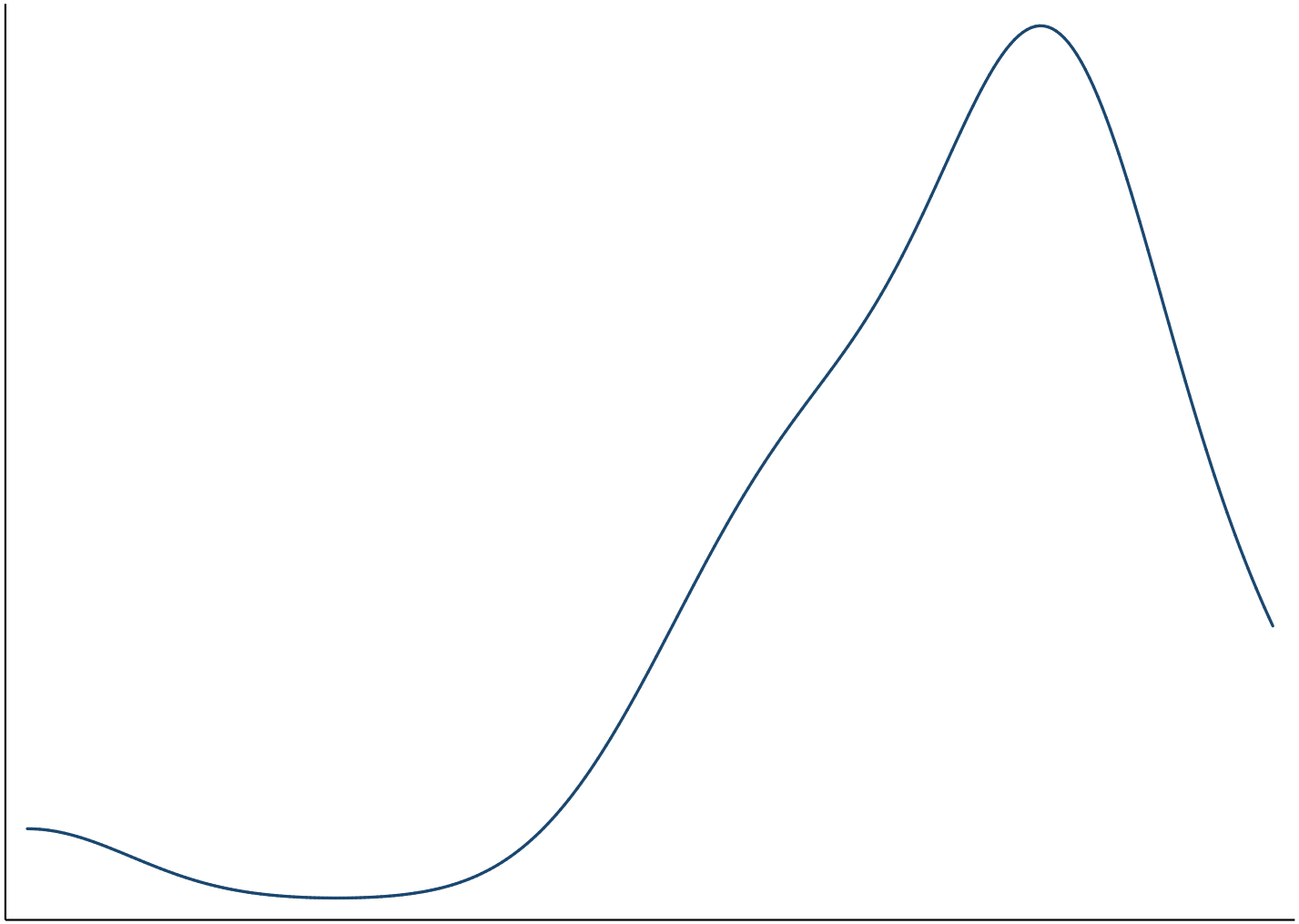
Reminder

- Due of Assignment #3: Sep 22, 2020 12:00 PM
- Midterm: Oct 1

Contents

- Bivariate Distributions
- Conditional Distributions
- Assignment 2

Illustration



Joint Distribution/Mass/Density

Definition

Suppose that random variables (X, Y) have support $S_X \times S_Y$, then

- Joint distribution function $F(x, y) = P(X \leq x, Y \leq y)$
- Joint mass $P(x, y) = P(X = x, Y = y)$
- Joint density $f(x, y)$ s.t.

$$F(x, y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v) dv du$$

Note that if (X, Y) are continuous, then by Leibniz's rule, $\frac{\partial}{\partial y} F(x, y) = \int_{-\infty}^{x} f(u, y) dv$ and

$$\frac{\partial^2}{\partial x \partial y} F(x, y) = f(x, y) = \frac{\partial^2}{\partial x \partial y} P(X \leq x, Y \leq y)$$

Joint Distribution/Mass/Density (Example)

- Bivariate normal distribution $(X_1, X_2) \sim N(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$ or

$$N(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \quad \text{where } \boldsymbol{\mu} = \begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$$

- PDF:

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2\left(1-\rho^2\right)} \left[\frac{\left(x-\mu_X\right)^2}{\sigma_X^2} + \frac{\left(y-\mu_Y\right)^2}{\sigma_Y^2} - \frac{2\rho\left(x-\mu_X\right)\left(y-\mu_Y\right)}{\sigma_X\sigma_Y} \right] \right)$$

or

$$f_{\mathbf{X}}\left(x_1, \ldots, x_k\right) = \frac{\exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)}{\sqrt{(2\pi)^k det(\boldsymbol{\Sigma})}}$$

Joint Distribution/Mass/Density (Example)

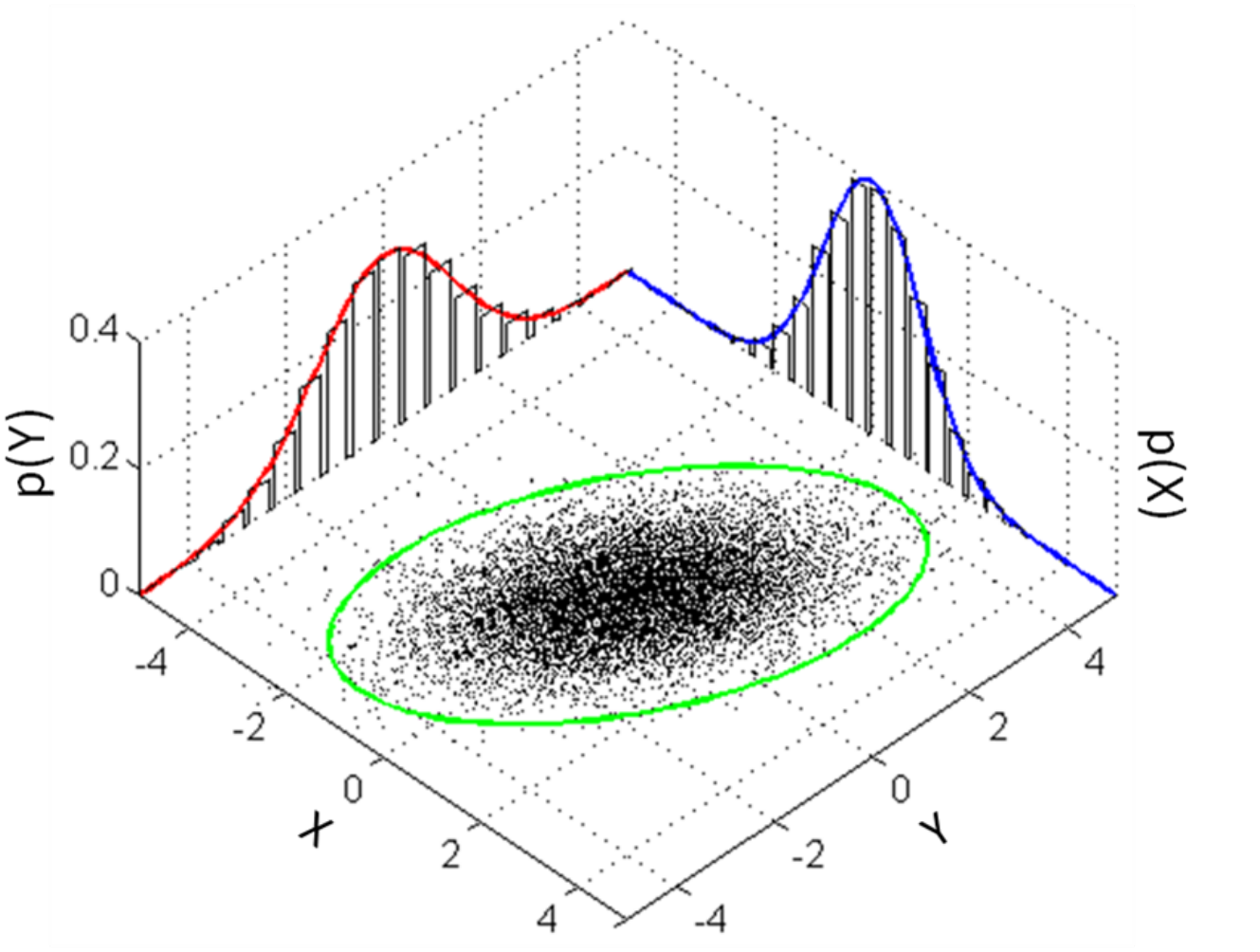


Figure: $\mu = (0, 0)$ and $\Sigma = \begin{pmatrix} 1 & 3/5 \\ 3/5 & 2 \end{pmatrix}$ (From Wikipedia)

Marginal Distribution/Mass/Density

Definition

Suppose that random variables (X, Y) have support $S_X \times S_Y$ with $F_{X,Y}(x, y)$, then

- Marginal distribution of X is $F(x) = P(X \leq x) = P(X \leq x, Y \in S_Y)$

$$= \int_{-\infty}^{x} \int_{S_Y} f_{X,Y}(u, v) dv du$$

- Mass $P(X = x) = P(X = x, Y \in S_Y)$

$$= \sum_{y \in S_Y} P(X = x, Y = y)$$

- Marginal density $f_X(x) = \frac{\partial}{\partial x} F_x(x) = \int_{-\infty}^{\infty} f(x, v) dv$

Marginal Distribution/Mass/Density (Example)

- Suppose that $(X, Y) \sim N(0, 0, 1, 1, 0)$
- PDF:

$$f(x, y) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}(x^2 + y^2)\right)$$

- Marginal density

$$f_X(x) = \int_{-\infty}^{\infty} f(x, v)dv = \int_{-\infty}^{\infty} \frac{1}{2\pi} \exp\left(-\frac{1}{2}(x^2 + v^2)\right) dv$$

$$= \frac{1}{2\pi} e^{-\frac{x^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{v^2}{2}} dv = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{v^2}{2}} dv = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

Event of two R.V.s

- Random event: $\{X > Y\}$
- Suppose that $(X, Y) \sim F_{X,Y}$ where PDF is $f_{X,Y}(x, y)$, then

$$P(X > Y) = \int_{S_X} \int_{-\infty}^{x} f_{X,Y}(x, y) dy dx$$

- Example: $(X, Y) \sim U[0, 1] \times U[0, 1]$ and $f_{X,Y}(x, y) = 1$, then

$$P(X > Y) = \int_0^1 \int_0^x f_{X,Y}(x, y) dy dx = \int_0^1 \int_0^x 1 dy dx = \int_0^1 x dx = \frac{1}{2}$$

Expectation

Definition

Suppose that random variables (X, Y) have support $S_X \times S_Y$, then

- If (X, Y) are discrete R.V.s, then

$$\mathbb{E}[g(X, Y)] = \sum_{x \in S_x} \sum_{y \in S_y} g(x, y) P(X = x, Y = y)$$

- If (X, Y) are continuous R.V.s, then

$$\mathbb{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

Conditional Distribution/Mass/Density

Recall that $P(A|B) = P(AB)/P(B)$. Similarly, under some regularities,

Definition

- Conditional distribution function

$$F_{Y|X}(y|x) = P(Y \leq y|X = x) = \frac{\frac{\partial}{\partial x}F(x,y)}{f_X(x)}$$

- Conditional density function

$$f_{Y|X}(y|x) = \frac{\partial}{\partial y}F_{Y|X}(y|x) = \frac{f(x,y)}{f_X(x)}$$

To prove this, please see the lecture slides.

Independence

Recall that events A and B are independent $\Leftrightarrow P(AB) = P(A)P(B)$, then

Definition

Random variable X and Y are independent if and only if

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$$

As a result, we have X and Y are independent if and only if

$$F_{X,Y}(x,y) = F_X(x)F_Y(y) \quad \text{or} \quad f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

or

$$F_{Y|X}(y|x) = F_Y(y) \quad \text{or} \quad f_{Y|X}(y|x) = f_Y(y)$$

Independence (Contd)

Theorem

If random variables X and Y are independent, then for any functions $g(X)$ and $h(Y)$, we have $g(X)$ and $h(Y)$ are independent. (Proof is easy when $g(\cdot)$ and $h(\cdot)$ are invertible)

Therefore, independent X and Y

$$\Rightarrow \quad \mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$$

for all $g(\cdot), h(\cdot)$ s.t. expected value is finite.

Indicator Functions

There's a link between indicator function and probability function,

$$\mathbb{E}[\mathbb{1}\{A\}] = P(A) \quad \text{for any event } A$$

For example, if $g(X) = \mathbb{1}\{X \leq a\}$ and $h(Y) = \mathbb{1}\{Y \leq b\}$, then

$$\mathbb{E}[g(X)] = \mathbb{E}[\mathbb{1}\{X \leq a\}] = P(X \leq a)$$
$$\mathbb{E}[h(Y)] = \mathbb{E}[\mathbb{1}\{Y \leq b\}] = P(Y \leq b)$$

Therefore, since

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[\mathbb{1}\{X \leq a\}\mathbb{1}\{Y \leq b\}] = \mathbb{E}[\mathbb{1}\{X \leq a, Y \leq b\}],$$

if

$$\mathbb{E}[\mathbb{1}\{X \leq a, Y \leq b\}] = \mathbb{E}[\mathbb{1}\{X \leq a\}]\mathbb{E}[\mathbb{1}\{Y \leq b\}]$$

then X and Y are independent.

Some Remarks about Assign 2

- Absolute value

$$|X - \mu| \leq M \quad \Leftrightarrow \quad -M \leq X - \mu \leq M$$

where $M > 0$, so

$$P(|X - \mu| \leq M) = P(-M \leq X - \mu \leq M)$$

- Derivative?

$$\frac{d|x|}{dx}$$

- Show $f(x)$ is symmetric at $\mu = \mathbb{E}[X] \Rightarrow$ Assume $f(x)$ is symmetric at μ?

Assign 2 - Question 5

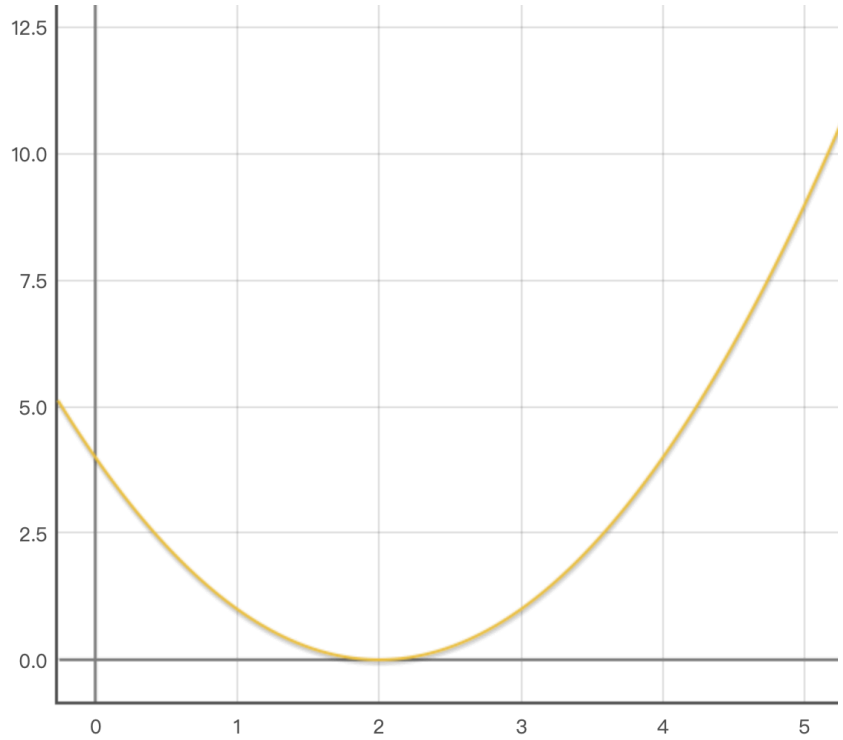
Question

Let $X \sim f(x) = \frac{1}{\lambda} e^{-\frac{x}{\lambda}}$ where $\lambda > 0$ and $X \geq 0$. Consider the transformation $Y = (X - \lambda)^2$. What's the support S_Y of Y? Derive $f_Y(y)$.

One of the solutions (and maybe the simplest) is in the answer key. Please firstly make sure you understand that one. Here I provide another approach using the theorem we introduced in previous recitation for your reference.

Assign 2 - Question 5 (Solution)

To apply the theorem, first we need to find the monotone part on support $S_X = [0, +\infty)$.



Based on the figure of $Y = g(X) = (X - \lambda)^2$ (here we let $\lambda = 2$ to illustrate it), we know $g(\cdot)$ is decreasing on $[0, \lambda)$ and increasing on $(\lambda, +\infty)$

Assign 2 - Question 5 (Solution)

Then we can define partitions $A_0 = \{\lambda\}$, $A_1 = [0, \lambda)$ and $A_2 = (\lambda, +\infty)$, then, since $y = (x - \lambda)^2 \in [0, +\infty) \iff \pm\sqrt{y} = x - \lambda$

- When $x \in A_1$, $y = g_1(x) = (x - \lambda)^2$ implies $x = g_1^{-1}(y) = \lambda - \sqrt{y}$ and $S_{Y1} = [0, \lambda^2]$
- When $x \in A_2$, $y = g_2(x) = (x - \lambda)^2$ implies $x = g_2^{-1}(y) = \lambda + \sqrt{y}$ and $S_{Y2} = [0, +\infty)$

In addition, we can get

$$\left|\frac{\partial}{\partial y} g_1^{-1}(y)\right| = \left|\frac{\partial}{\partial y} g_2^{-1}(y)\right| = \frac{1}{2\sqrt{y}}$$

Therefore, by theorem,

$$f_Y(y) = \sum_{i=1}^{2} f_X(g_i^{-1}(y)) \left|\frac{\partial}{\partial y} g_i^{-1}(y)\right| \mathbb{1}\{y \in S_{Yi}\}$$

Assign 2 - Question 5 (Solution)

(Continued)

$$
\begin{aligned}
f_Y(y) &= \sum_{i=1}^{2} f_X(g_i^{-1}(y)) \left| \frac{\partial}{\partial y} g_i^{-1}(y) \right| \mathbb{1}\{y \in S_{Yi}\} \\
&= \frac{1}{\lambda} e^{-\frac{1}{\lambda}(\lambda - \sqrt{y})} \frac{1}{2\sqrt{y}} \mathbb{1}\{0 \le y \le \lambda^2\} \\
&\quad + \frac{1}{\lambda} e^{-\frac{1}{\lambda}(\lambda + \sqrt{y})} \frac{1}{2\sqrt{y}} \mathbb{1}\{0 \le y\} \\
&= \frac{1}{2\lambda\sqrt{y}} \left(e^{-1+\frac{\sqrt{y}}{\lambda}} \mathbb{1}\{0 \le y \le \lambda^2\} + e^{-1-\frac{\sqrt{y}}{\lambda}} \mathbb{1}\{y \ge 0\} \right)
\end{aligned}
$$

which is the same as the one in answer key.